POWER BI DASHBOARD PORTFOLIO

Eric Kwamena Asibu Yartey

Table of Contents

INTRODUCTION

This Power BI dashboard portfolio showcases three distinct projects, demonstrating expertise in data analysis, visualisation, and strategic reporting across the domains of public health, humanitarian aid, and business intelligence

1. **COVID-19 Vaccine Accountability Dashboard (Ghana EPI)**: Developed for the Expanded Programme on Immunisation (EPI) in Ghana, this dashboard project aimed to enhance oversight of COVID-19 vaccine accountability by tracking unusable, wasted, and expired vaccines. By utilising data from Kobo Toolbox, the dashboard facilitated real-time reporting, enabling timely incineration planning, reducing reporting delays, and promoting transparency for donors.

2. **Syria Intervention Dashboard (PEOPLE IN NEED)**: Developed for the international NGO, People in Need, this dashboard project monitored the effectiveness of humanitarian interventions in Syria. It tracked beneficiaries by location and age, the aid provided, and dependency ratios, facilitating data-driven targeting, enhancing reporting efficiency, and assisting in gap analysis across humanitarian sectors.

3. **MNCH Commodity Stock Dashboard (Chemonics International)**: Developed for Chemonics International, this dashboard project focused on monitoring the availability and stock-out rates of Maternal, Newborn, and Child Health (MNCH) commodities in underserved regions of Ghana. It enabled real-time monitoring, facilitated the redistribution of supplies, informed procurement plans for the Ministry of Health, and assisted in securing additional donor funding.

4. **PoD Verification and 3PL Capacity Assessment Dashboard (GHSC-PSM)**: Developed for GHSC-PSM, this dashboard project monitored the performance and capacity of third-party logistics (3PL) partners within Ghana's health supply chain. The dashboard validated transporter claims, facilitated route-level performance monitoring, identified gaps in cold chain capacity, and enhanced the utilisation of data within the national supply chain.

5. **Sales Intelligence Dashboard (SNOWFLAKE)**: This dashboard project utilised Snowflake's sales database to conduct a comprehensive analysis of sales, including cumulative sales, product sales, and profit margins. It incorporated cohort analysis to gain insights into customer retention and employed time-series forecasting for future profit predictions. As a result, the project led to optimised inventory management, enhanced marketing campaigns, and increased visibility for executives.

Each project demonstrates the effective use of Power BI for data-driven decision-making, enhancing operational efficiency, accountability, and strategic planning across various organisational contexts.

Project: COVID-19 Vaccine Accountability Dashboard (Ghana EPI)

Background

Ghana's Expanded Programme on Immunization (EPI) is responsible for COVID-19 vaccine accountability and monitoring. This involves tracking the usage of COVID-19 vaccines, ensuring quality, and evaluating program performance to maximize impact and ensure proper resource allocation. This process includes monitoring the COVID-19 vaccine supply chains, tracking immunization coverage, and assessing the effectiveness of immunization strategies. Key aspects of this initiative include monitoring vaccine wastage, managing the cold chain, and addressing adverse events following immunization (AEFI).

The objective of this dashboard project was to enhance oversight of COVID-19 vaccine accountability by reporting and analysis:

- Unusable vaccines (damaged, broken vials, improperly handled)
- Wasted vaccines (open vial wastage, unused doses after sessions)
- Expired vaccines (past manufacturer expiry date)

This was critical for:

- Ensuring proper waste classification and disposal (e.g., incineration)
- Supporting cold chain compliance and quality control
- Aiding donor transparency and reporting
- Informing redistribution and resupply decisions

Approach

1. **Data Collection & Sources**
    - Data was submitted weekly from all COVID vaccination centres via Kobo Toolbox. Kobo Toolbox data connected to Power BI via Kobo API.

Key fields captured:

- Region, District, Facility Name
- Vaccine Type (e.g., AstraZeneca, Moderna, Pfizer, J&J,Sputnik_V)
- Batch Number
- Number of vials: Used, Usable, Missing, Expired, Broken, Spoilt
- Reason for wastage
- Date of reporting

2. **Data Modelling**
   - Data extracted from Kobo Toolbox.
   - Cleaned and structured using Power Query
   - Model created:
     - Fact_vaccines – contains all vaccine records
     - Dim_Facility, Dim_Location, Dim_VaccineList
   - Created calculated measures using DAX (e.g., ttl_Broken_Vials, ttl_Used_Vials, ttl_Expired_Vials etc.)

3. **Dashboard Features**
   - **KPIs**: Vaccine Districts, Vaccine Centres, Issued_Vials, Used_Vials, Usable_Vials, Expired_Vials, Missing_Vials, Broken_Vials, Spoilt_Vials.
   - **Visuals**:
     - Table showing name of vaccine and the quantity issued, expired, missing, broken, spoilt, all in vials.
     - Funnel showing the percentages of usable, used, expired, missing, broken, and spoilt vials.
     - Clustered column chart showing total vials issued by vaccine type versus vials usable, used, missing, expired, broken, and spoilt.
     - Tables showing vaccine counts in vials versus in doses for usable, used, expired, missing, broken, and spoilt.

        - Clustered column chart showing percentage of issued vials by region, percentage of usable vials by region, percentage of expired vials by region, percentage of used vials by region, percentage of broken vials by region, percentage of spoilt vials by region, and percentage of missing vials by region.
    - **Interactivity:**
        - **Slicer**: Filter by Vaccine Type, Filter by Region & District.

4. **Deployment**
    - Published to Power BI Service.
    - Scheduled data refresh weekly.
    - Shared with Vaccine Program Leads and District /Regional Officers via Power BI Service

<u>Impact</u>

- Centralized real-time reporting of unusable and expired vaccines nationwide.
- Enabled timely planning of incineration logistics by regional cold stores.
- Reduced reporting delays compared to legacy manual summaries.
- Supported accountability in reporting to COVAX and development partners.

238
Vaccine Districts

2,703
Vaccine Centres

2,371,871
Issued_Vials

1,992,993
Used_Vials

319,470
Useable_Vials

55,708
Expired_Vials

1,955
Missing_Vials

728
Broken_Vials

1,017
Spoilt_Vials

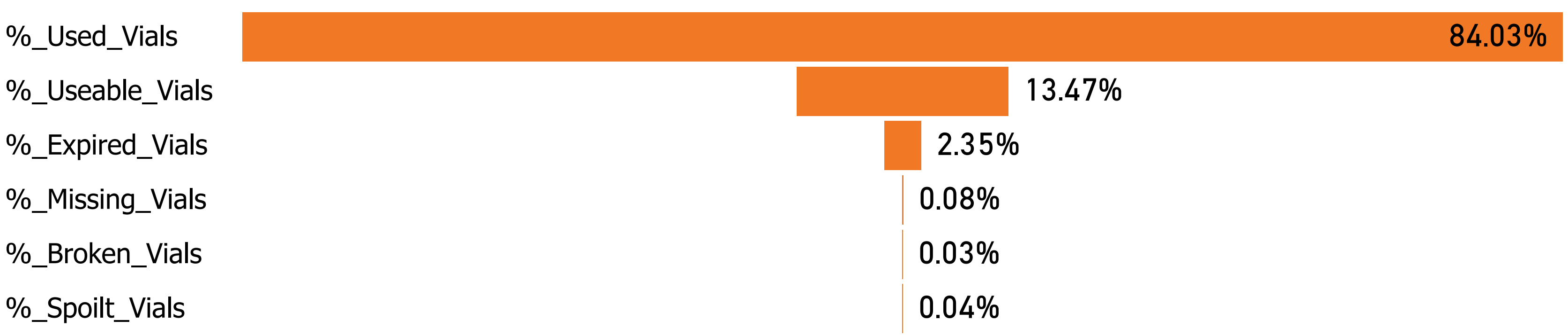
COVID-19 VACCINE ACCOUNTABILITY

VaccineName	Issued_Vials	Used_Vials	Useable_Vials	Spoilt_Vials	Missing_Vials	Expired_Vials	Broken_Vials
AstraZeneca	1,248,256	1,089,159	122,619	22	961	34,955	540
Johnson_Johnson	431,864	328,924	99,182	13	126	3,557	62
Moderna	105,226	95,047	6,560	5	105	3,455	54
Pfizer	576,468	470,985	90,159	977	763	13,514	70
Sputnik_V	10,057	8,878	950			227	2
Total	**2,371,871**	**1,992,993**	**319,470**	**1,017**	**1,955**	**55,708**	**728**

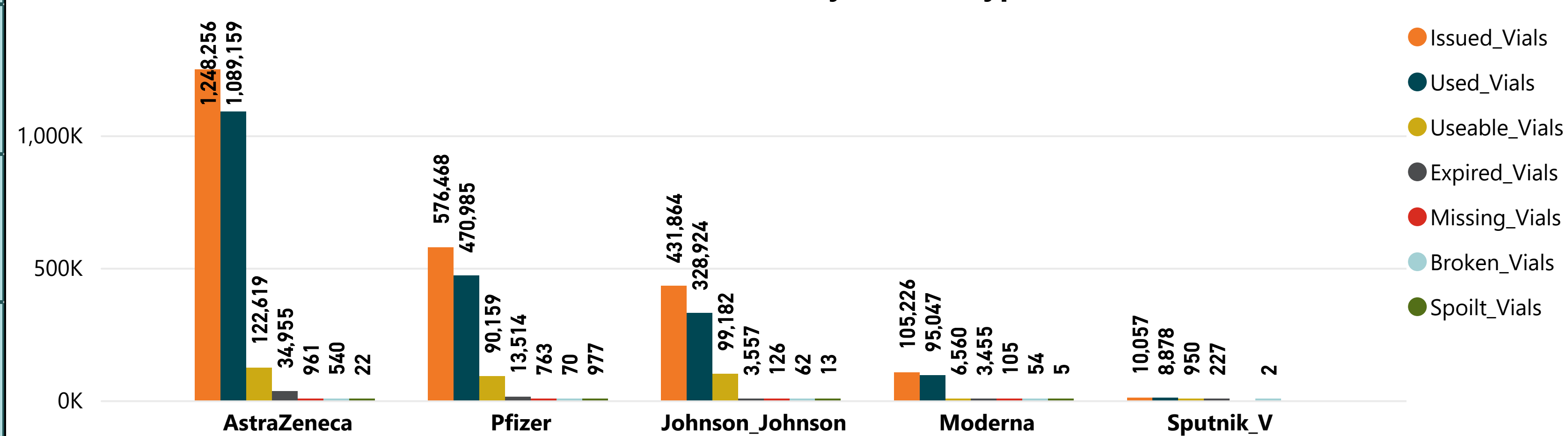
Vaccine List

- AstraZeneca
- Johnson_Johnson
- Moderna
- Pfizer
- Sputnik_V

Perspectives in Percentages



Total Vials Issued by Vaccine Type



Region List

- Ahafo
- Ashanti
- Bono
- Bono East
- Central
- Eastern
- Greater Accra
- North East
- Northern
- Oti
- Savannah
- Upper East
- Upper West
- Volta
- Western
- Western North

Vaccine Count (in Vials)

VaccineName	Issued_Vials	Used_Vials	Useable_Vials	Spoilt_Vials	Missing_Vials	Expired_Vials	Broken_Vials
AstraZeneca	1,248,256	1,089,159	122,619	22	961	34,955	540
Johnson_Johnson	431,864	328,924	99,182	13	126	3,557	62
Moderna	105,226	95,047	6,560	5	105	3,455	54
Pfizer	576,468	470,985	90,159	977	763	13,514	70
Sputnik_V	10,057	8,878	950			227	2
Total	**2,371,871**	**1,992,993**	**319,470**	**1,017**	**1,955**	**55,708**	**728**

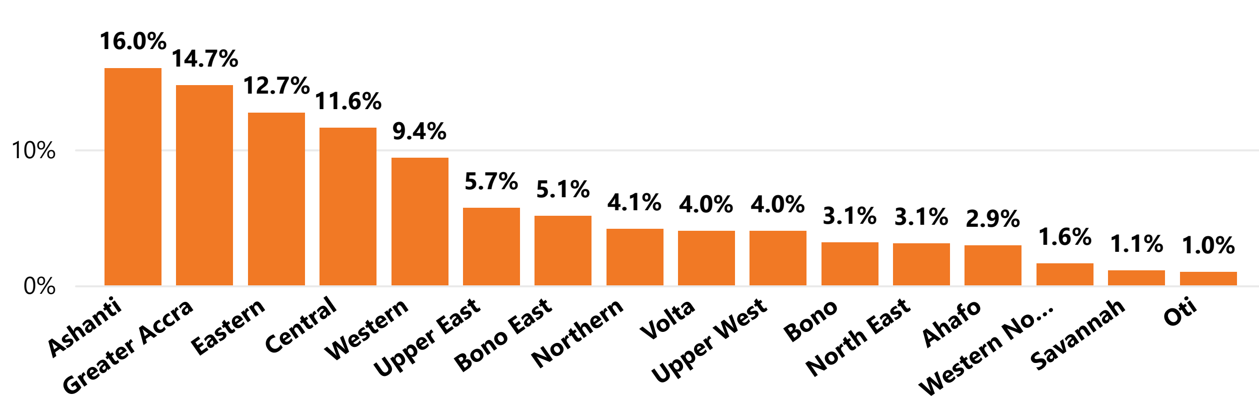
Vaccine Count (in Doses)

VaccineName	Doses per Vial	Doses Issued	Doses Used	Doses Useable	Doses Spoilt	Doses Missing	Doses Expired	Doses Broken
AstraZeneca	10	12,482,560	10,891,590	1,226,190	220	9,610	349,550	5,400
Johnson_Johnson	5	2,159,320	1,644,620	495,910	65	630	17,785	310
Moderna	14	1,473,164	1,330,658	91,840	70	1,470	48,370	756
Pfizer	6	3,458,808	2,825,910	540,954	5,862	4,578	81,084	420
Sputnik_V	6	60,342	53,268	5,700			1,362	12
Total		**19,634,194**	**16,746,046**	**2,360,594**	**6,217**	**16,288**	**498,151**	**6,898**

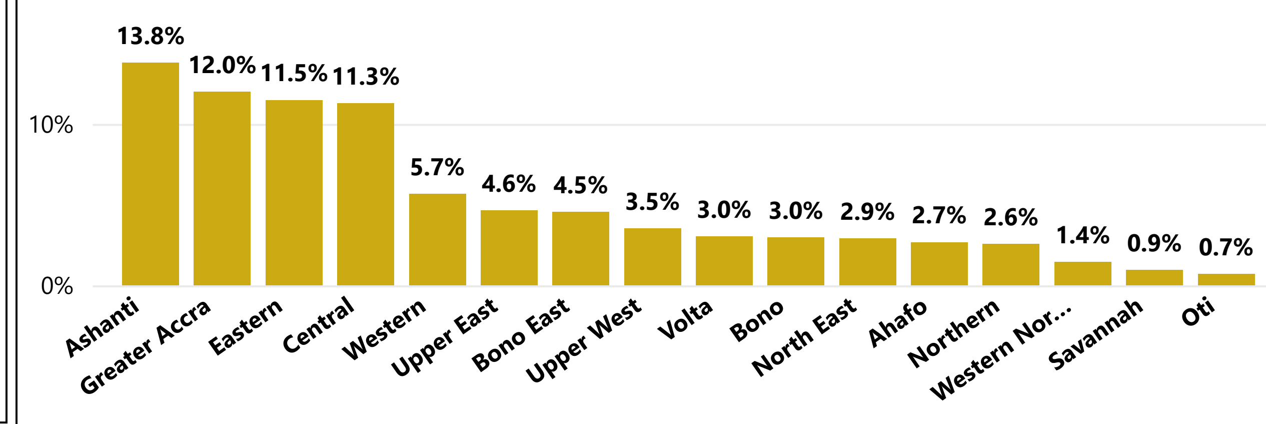
Region List

- Ahafo
- Ashanti
- Bono
- Bono East
- Central
- Eastern
- Greater Accra
- North East
- Northern
- Oti
- Savannah
- Upper East
- Upper West
- Volta
- Western
- Western North

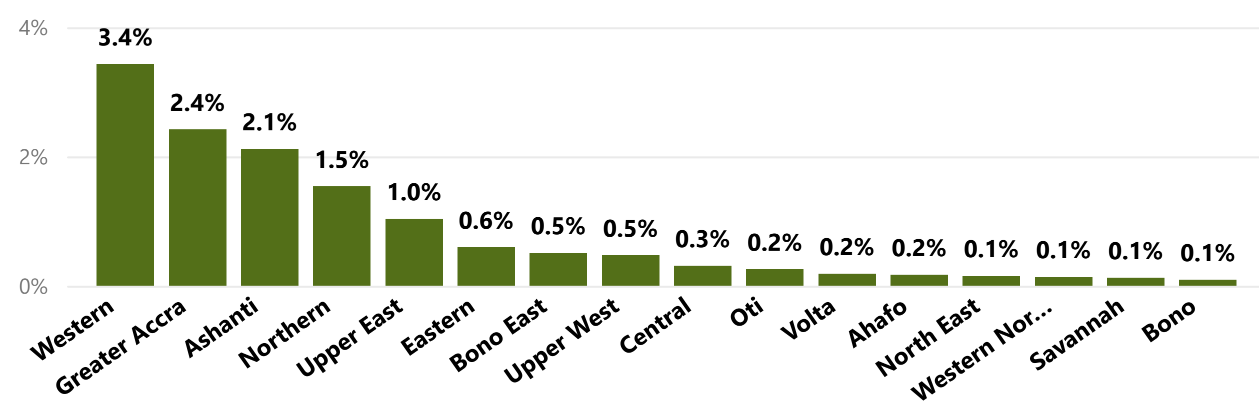
Percentage of Issued Vials by Region



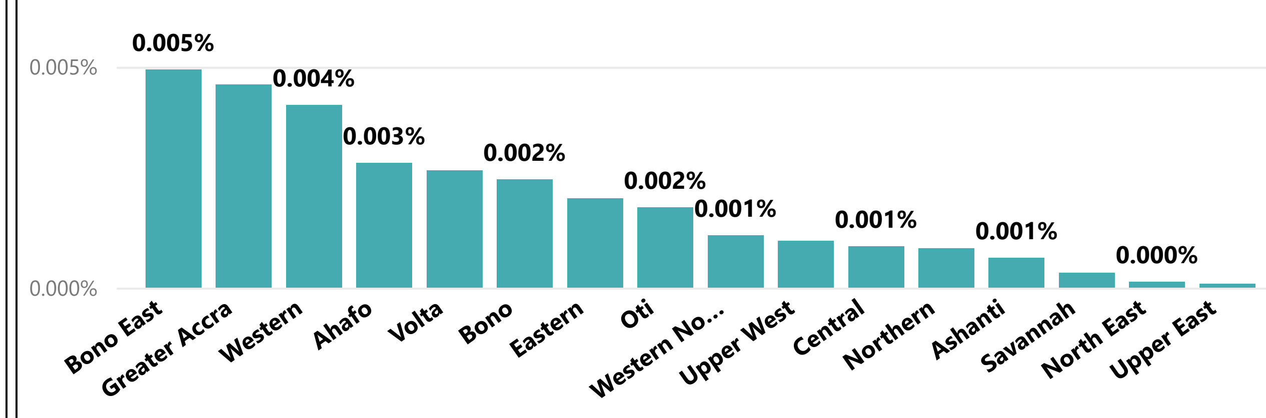
Percentage of Used Vials by Region



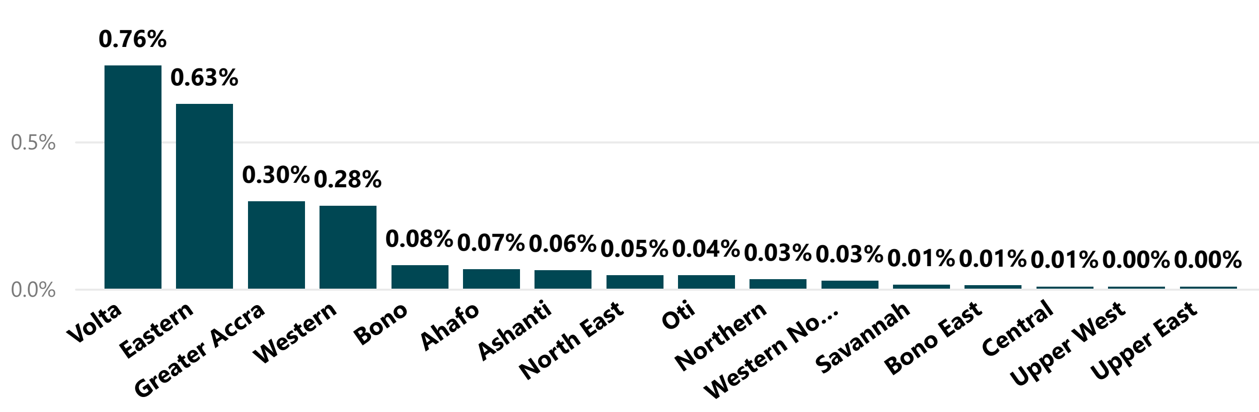
Percentage of Useable Vials by Region



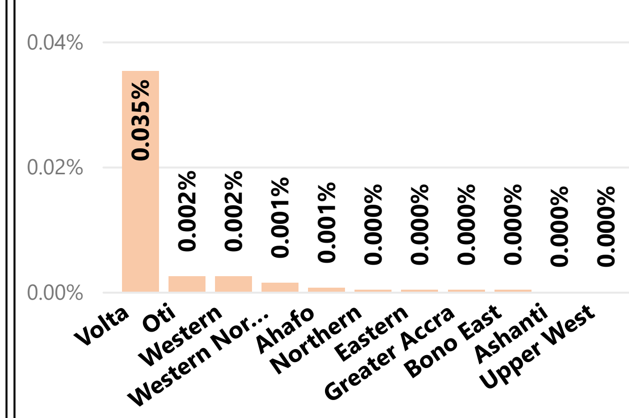
Percentage of Broken Vials by Region



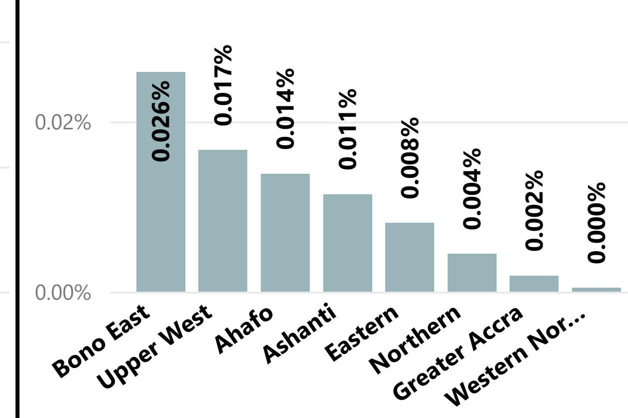
Percentage of Expired Vials by Regionon



Percentage of Spoilt Vials by Region



Percentage of Missing Vials by Region



Project: Syria Intervention Dashboard (PEOPLE IN NEED)

Background

People In Need, an international NGO that offers humanitarian, development, and human rights support in over 40 countries worldwide, aimed to monitor and evaluate the effectiveness of its interventions for conflict-affected populations in Syria. The organization provides food aid, repairs infrastructure and water supply network, offers shelter to vulnerable individuals, distributes hygiene kits, and assists displaced families in enduring harsh winters. The objective of this project was to equip regional program managers with a real-time, interactive dashboard to track:

- the total number of beneficiaries per location (area).
- the age breakdown of beneficiaries.
- the amount of aid provided to beneficiaries.
- the average dependency ratio in different locations.
- the average income of groups coming from different locations on different dates.
- the age profile in different vulnerability criteria categories.

Approach

1. **Data Collection & Sources**
    - Monthly reports from field officers collected via Kobo Toolbox. Kobo Toolbox data connected to Power BI via Kobo API.
    - Population needs/targets from inter-agency assessments.
    - Data cleaned and pre-processed in Power Query.

2. **Data Modeling**
    - Developed a star schema model: Fact table for Household size and dimension tables for Household_Criteria, Support_Status, Socio_Economic_Criteria, Aid_Donation etc.

- Created calculated measures using DAX (e.g., Total_Food_Kits_Delivered, Total_Hygiene_Kits_Delivered, %_of_Aid_Amount, %_of_Beneficiaries, %_of_Male, %_of_Female etc.).

3. **Dashboard Features**
   - **Cards**: Total beneficiaries, Total Food Kits, Total Hygiene Kits, Total Aid Amount.
   - **Visuals**:
     - Bar charts showing Beneficiaries by Area, Beneficiaries by Age group, and Average Income of Groups by Area and Date.
     - Pie chart for gender information, and aid amount to residents & IDPs.
     - Funnel showing vulnerability categories
     - Tables: Beneficiaries by Area, Age breakdown of Beneficiaries, Aid Amount to Residents & IDPs, Average Dependency Ratio, Vulnerability Categories by Area.
   - **Interactivity:**
     - **Slicer**: Filter by Area, Filter by Year/Month.

4. **Deployment**
   - Published to Power BI Service.
   - Scheduled data refresh weekly.
   - Shared with PIN management and PIN Syria local team via workspace access and embedded in internal SharePoint site.

<u>Impact</u>

- Enabled data-driven targeting, redirecting supplies to under-reached communities.
- Improved reporting efficiency for cluster coordination and donor updates.
- Facilitated gap analysis across humanitarian sectors.

Syria Intervention Dashboard

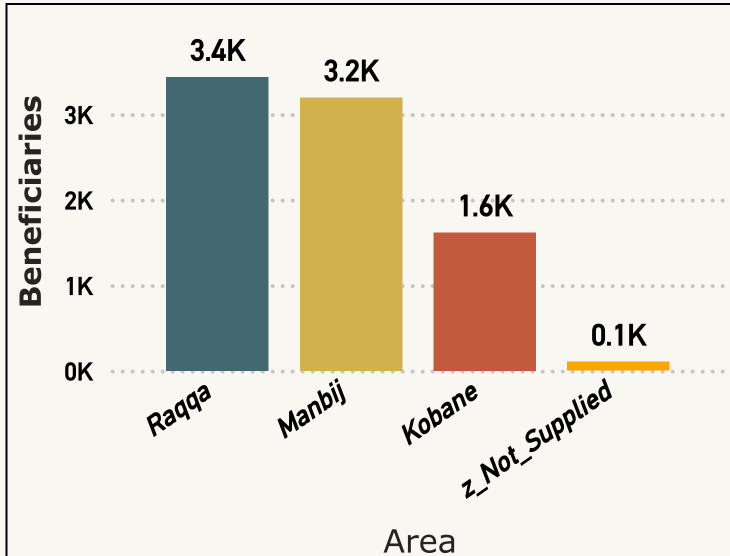
Total Beneficiaries	Total Food Kits	Total Hygiene Kits	Total Aid Amount
8,350	25.05K	14.76K	€ 46.32M

Filter By Area

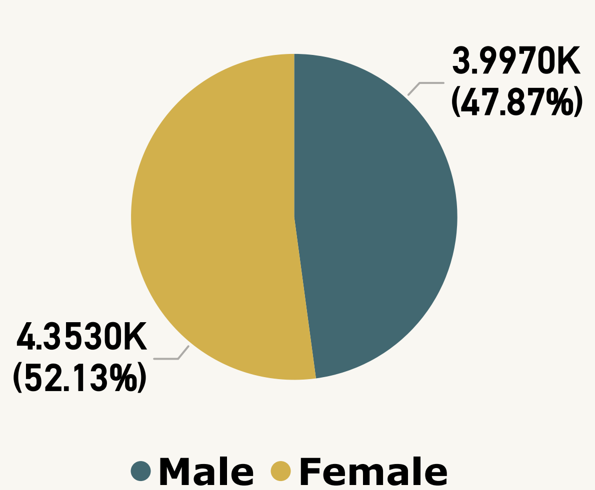
- Kobane
- Manbij
- Raqqa
- z_Not_Supplied

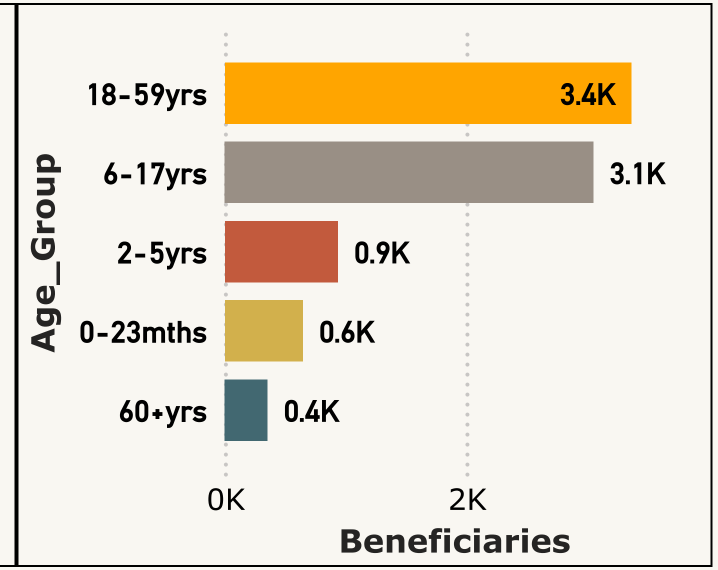
Filter By Year/Mth

- 2021
  - January
  - February
  - March
  - April
  - May
  - June
  - July
  - August
  - September
  - October
  - November
  - December



Gender Information





Number of Beneficiaries by Area

Area	Beneficiaries	%_of_Benefficiaries
Kobane	1,616	19.35%
Manbij	3,195	38.26%
Raqqa	3,435	41.14%
z_Not_Supplied	104	1.25%
Total	**8,350**	**100.00%**

Age Breakdown of Beneficiaries

Age_Group	Beneficiaries	%_of_Beneficiaries	Males	%_of_Males	No_of_Females	%_of_Females
0-23mths	645	7.72%	340	4.07%	305	3.65%
18-59yrs	3,366	40.31%	1,486	17.80%	1,880	22.51%
2-5yrs	935	11.20%	490	5.87%	445	5.33%
60+yrs	352	4.22%	146	1.75%	206	2.47%
6-17yrs	3,052	36.55%	1,535	18.38%	1,517	18.17%
Total	**8,350**	**100.00%**	**3,997**	**47.87%**	**4,353**	**52.13%**

Syria Intervention Dashboard

Average Dependency Ratio

Area	Avg_DependencyRatio
Kobane	2.40
Manbij	2.34
Raqqa	2.39
z_Not_Supplied	2.35

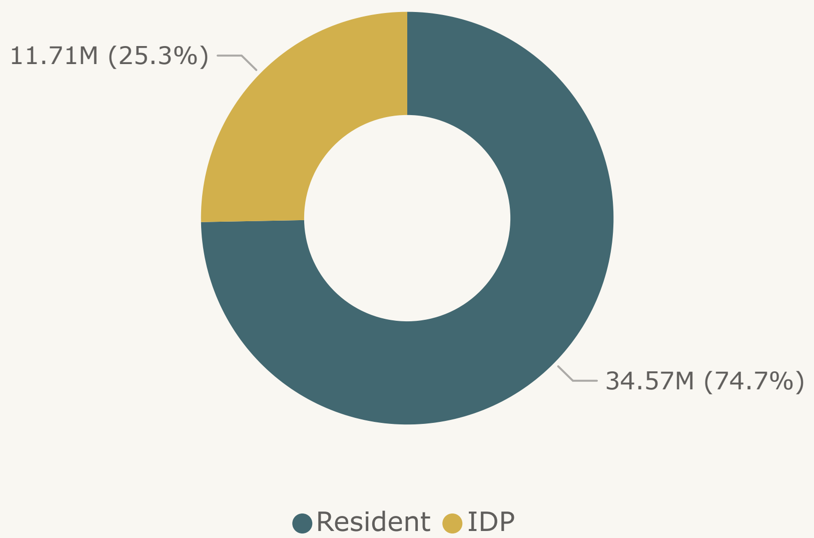
Filter By Area

- Kobane
- Manbij
- Raqqa
- z_Not_Supplied

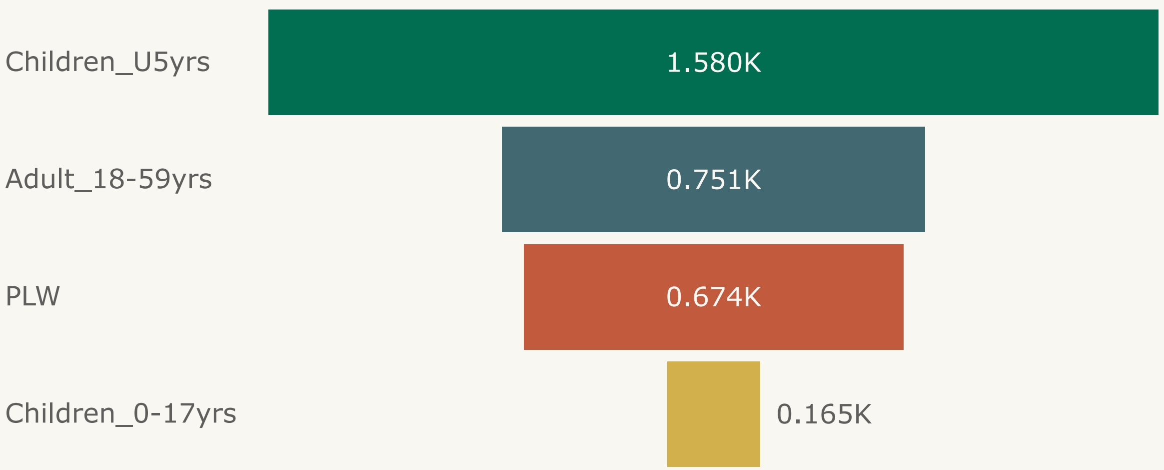
Filter By Year/Mth

- 2021
  - January
  - February
  - March
  - April
  - May
  - June
  - July
  - August
  - September
  - October
  - November
  - December

Aid Amount to Residents & IDPs



Vulnerability Categories



Aid Amount to Residents/IDPs

VC_Resident_Status	Aid_Amount	%_of_Aid_Amount
IDP	11,710,500.00	25.30%
Resident	34,574,052.00	74.70%
Total	**46,284,552.00**	**100.00%**

Vulnerability Categories By Area

Area	Children_0-17yrs	Children_U5yrs	PLW	Adult_18-59yrs	Total_Members
Kobane	29	317	136	144	626
Manbij	62	587	268	338	1,255
Raqqa	71	663	262	260	1,256
z_Not_Supplied	3	13	8	9	33
Total	**165**	**1,580**	**674**	**751**	**3,170**

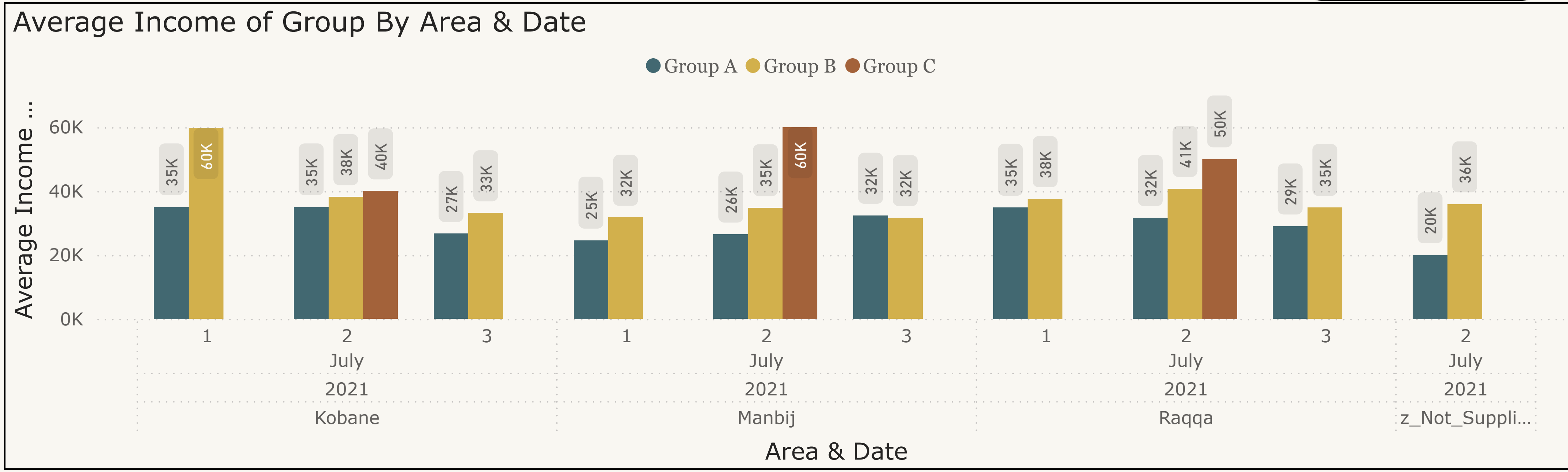
Syria Intervention Dashboard

Filter By Area

- Kobane
- Manbij
- Raqqa
- z_Not_Supplied

Filter By Year/Mth

- 2021
  - January
  - February
  - March
  - April
  - May
  - June
  - July
  - August
  - September
  - October
  - November
  - December



Average Income of Group By Area and Date

Year	Month	Day	Area	Group	Avg_Monthly_Income
2021	July	1	Kobane	Group A	35,000.00
2021	July	2	Kobane	Group A	35,000.00
2021	July	3	Kobane	Group A	26,666.67
2021	July	1	Kobane	Group B	59,703.13
2021	July	2	Kobane	Group B	38,179.78
2021	July	3	Kobane	Group B	33,125.00
2021	July	2	Kobane	Group C	40,000.00
2021	July	1	Manbij	Group A	24,588.24

Project: MNCH Commodity Stock Dashboard (Chemonics International)

Background

In Ghana, the availability of Maternal, Newborn, and Child Health (MNCH) commodities and stock-out rates vary across different types of healthcare facilities. According to a supply chain assessment, while stock-out rates for many MNCH commodities are generally low, Oral Rehydration Salts (ORS) have exhibited a relatively higher stock-out rate of 27% at wholesale outlets. Retail pharmacies and wholesalers are less likely to manage injectable products such as gentamicin, magnesium sulfate, and oxytocin. Chemonics International, an international NGO implementing the USAID Global Health Supply Chain project in Ghana, aims to monitor the availability and stock-out rates of selected MNCH commodities—ORS, Zinc Tablets, Amoxicillin Suspension (125 mg/5 ml), Oxytocin Injection (10 mg/ml), Gentamicin (40 mg/ml), Chlorhexidine Digluconate (7.1% Gel), Magnesium Sulfate (50%), and Hydralazine HCl (20 mg/ml Injection)—at some healthcare facilities in the four most underserved regions of Ghana: Northern, North East, Upper East, and Upper West.

The objective of this project was to:

- Track the availability and stock-outs of critical MNCH items.
- Identify trends and supply gaps in the selected healthcare facilities across Northern Ghana.
- Support timely resupply decisions and evidence-based planning.

Approach

1. **Data Collection & Sources**
    - The data was collected weekly from 93 primary healthcare facilities via Kobo Toolbox. Kobo Toolbox data connected to Power BI via Kobo API.

- Selected MNCH Commodities included: *ORS, Zinc Tablets, Amoxicillin Suspension (125 mg/5 ml), Oxytocin Injection (10 mg/ml), Gentamicin (40 mg/ml), Chlorhexidine Digluconate (7.1% Gel), Magnesium Sulfate (50%), and Hydralazine HCl (20 mg/ml Injection).*
- Key fields collected:
    - Facility Name, District, Region
    - Commodity Name
    - Opening Balance, Quantity Received, Quantity Issued, Closing Balance
    - Reported Stock-Out (Yes/No), Days Out of Stock

2. **Data Preparation & Modelling**
    - Cleaned and merged in Power Query.
    - Developed a star schema model: Fact table logistics linked to dimension tables: region, product, facility, date_dim.
    - Created calculated columns in the date_dim table using DAX (e.g. FirstDateOfWeek, LastDateOfWeek, MonthYear, WeekInMonth, WeekOfDate, etc)
    - Created calculated measures using DAX (e.g., Facility_with_MoS, Facility_with_Stockout, SoH, MoS, Last3-12MthsCons etc.)

3. **Dashboard Features**
    - **Visuals**:
        - Matrix showing availability of selected MNCH commodities as well months of stock for each commodity with drill-throughs for district-specific details.
        - Line charts to show trend and compare percentage of facilities running at stockout of selected MNCH commodities on monthly basis.
        - Stacked bar charts to compare percentage of facilities with months of stocks greater than zero for the selected MNCH commodities on monthly basis.

- **Interactivity:**
    - **Slicer**: Filter by Year/Month/Weeks, Filter by District/Facility, and Filter by Product.
    - **Drill-Throughs**: Applied on Matrix visuals for district-specific details.

4. **Deployment**
    - Published to Power BI Service.
    - Scheduled data refresh weekly.
    - Shared with MNCH program leads and Technical Directors via Power BI service.

Impact

- Enabled real-time monitoring of commodity availability in hard-to-reach regions.
- Facilitated quick redistribution of supplies from overstocked to understocked facilities.
- Informed the Ministry of Health's quarterly procurement and distribution plans.
- Provided evidence to secure additional donor funding for maternal health interventions.

Year/Month/Weeks

- (Blank)
- 2022
- 2023
- 2024
  - January
  - February
  - March
  - April
  - May
  - June
  - July
  - August
  - September

District/Facility

- (Blank)
- NorthEast
- Northern
- Upper East
- Upper West

MNCH Commodity Availability

Product Desc	Amoxicillin Suspension 125mg/5ml		Chlorhexidine Digluconate 7.1% Gel		Gentamicin 40mg/ml		Hydralazine HCl 20mg/ml Injection		Magnesium Sulphate 50%		ORS		Oxytocin Injection 10mg/ml		Zinc Tablet	
district	SoH	MoS	SoH	MoS	SoH	MoS	SoH	MoS	SoH	MoS	SoH	MoS	SoH	MoS	SoH	MoS
⊟ East Mamprusi																
Baptist Hospital	0	0.00	1110	11.10	2900	7.91	10	6.00	550	8.68	6900	31.85	3600	33.75	34500	20.70
Buzulangu CHPS	60	13.85	0	0.00	0	0.00	0	0.00	0	0.00	0	0.00	0	0.00	0	0.00
Dagbiriboari CHPS	123	10.85	66	9.00	0	0.00	0	0.00	0	0.00	0	0.00	0	0.00	0	0.00
Gambaga HC	0	0.00	15	IND	10	2.00	5	IND	5	3.00	30	3.00	10	2.00	20	1.50
Ghabugu CHPS	0	0.00									0	0.00			0	0.00
Langbinsi Health Center	120	6.00			0	0.00					19	3.35	510	15.30	400	1.22
Namangu CHPS	90	33.75	14	42.00	0	0.00	0	0.00	0	0.00	0	0.00	10	1.67	0	0.00
Samini CHPS	89	8.34	138	31.85	0	0.00					0	0.00	2	0.75	0	0.00
Wundua CHPS	80	1.55	0	0.00	0	0.00	0	0.00	0	0.00	0	0.00	0	0.00	0	0.00
Zaari CHPS	54	3.86	0	0.00							0	0.00			0	0.00
⊟ Mamprugu Moagduri																
Kubori Health Center	0	0.00	112	12.00	0	0.00	12	IND	10	IND	0	0.00	92	IND	0	0.00
Kubugu CHPS	0	0.00	8	3.43	0	0.00			0	0.00	0	0.00	9	4.50	0	0.00
Namoo CHPS	30	9.00	15	15.00	0	0.00					0	0.00			0	0.00
Sandema District Hospital	180	15.00	90	54.00	1500	7.99	68	9.71	28	3.50			290	6.85		
Tantala CHPS	0	0.00									0	0.00	0	0.00	0	0.00
Walewale District Hospital	672	4.94	950	28.50	100	0.75	300	180.00	300	3.00	1500	4.00	400	2.86	0	0.00
Yagaba CHPS	0	0.00	0	0.00	0	0.00	0	0.00	3	4.50	0	0.00	0	0.00	0	0.00
Yikpabongo CHPS									1	1.00			0	0.00		
Yizeizi Health Center	0	0.00	0	0.00	0	0.00	0	0.00	0	0.00	0	0.00	0	0.00	0	0.00

Legend **Overstock** **Adequate Stock** **Under Stock** **Stock Out**

Product

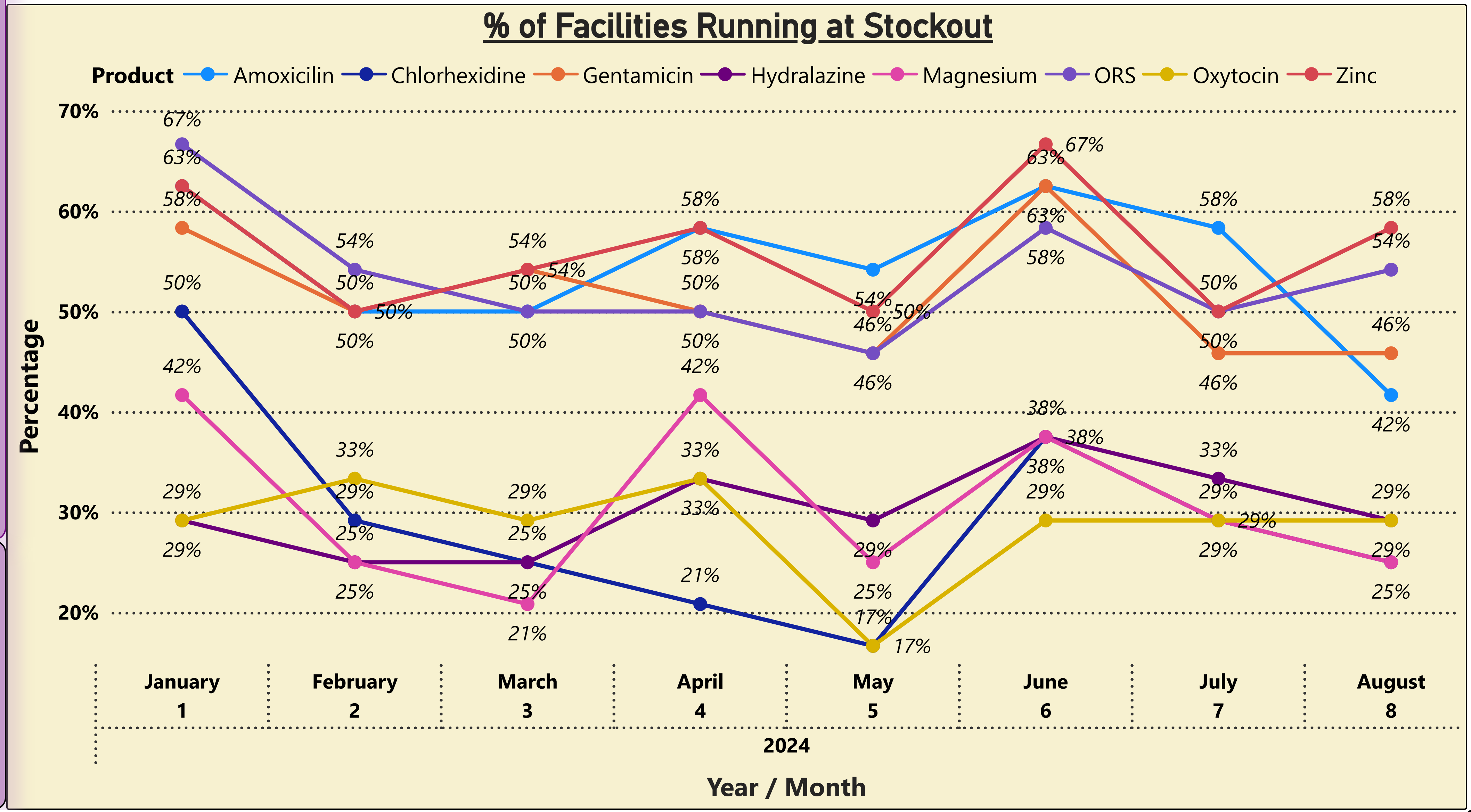
Select all	Amoxicillin Suspension 125mg/5ml (bot…	Chlorhexidine Digluconate 7.1% Gel (tube)	Gentamicin 40mg/ml (vial)	Hydralazine HCl 20mg/ml Injection (ampo…	Magnesium Sulphate 50% (vial)	ORS (sachet)	Oxytocin Injection 10mg/ml (vial)	Zinc Tablet (blister)

Year/Month/Weeks

- *(Blank)*
- *2022*
- *2023*
- *2024*
  - *January*
  - *February*
  - *March*
  - *April*
  - *May*
  - *June*
  - *July*
  - *August*
  - *September*
  - *October*

District/Facility

- *NorthEast*
- *Northern*
- *Upper East*
- *Upper West*



Product

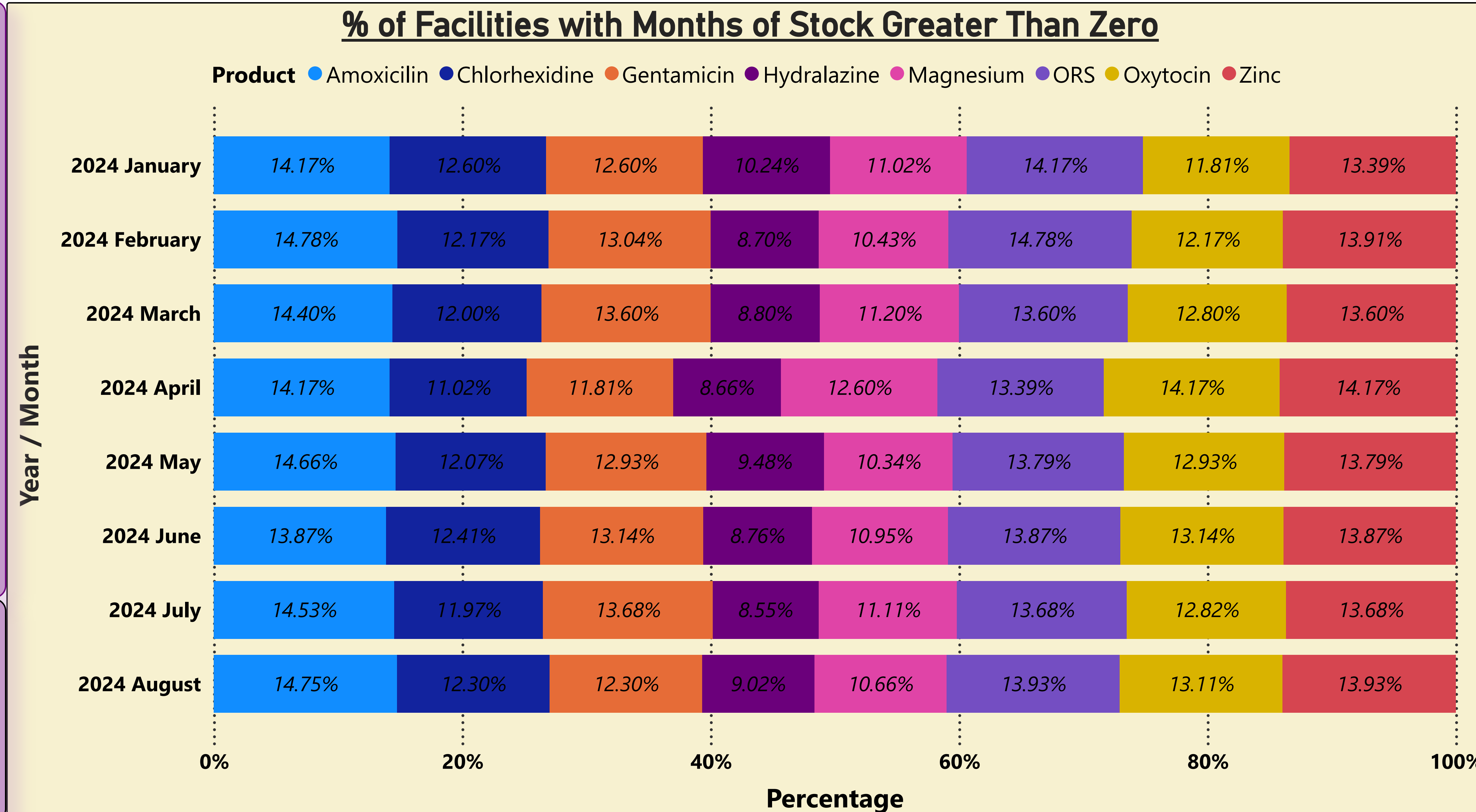
Select all	Amoxicillin Suspension 125mg/5ml (bot...	Chlorhexidine Digluconate 7.1% Gel (tube)	Gentamicin 40mg/ml (vial)	Hydralazine HCl 20mg/ml Injection (ampo...	Magnesium Sulphate 50% (vial)	ORS (sachet)	Oxytocin Injection 10mg/ml (vial)	Zinc Tablet (blister)

Year/Month/Weeks

- (Blank)
- 2022
- 2023
- 2024
  - January
  - February
  - March
  - April
  - May
  - June
  - July
  - August
  - September
  - October

District/Facility

- NorthEast
- Northern
- Upper East
- Upper West



Project: PoD Verification & 3PL Capacity Assessment Dashboard (GHSC-PSM)

Background

The USAID Global Health Supply Chain Program - Procurement and Supply Management (GHSC-PSM) is committed to ensuring a continuous supply of health commodities to support U.S. Government-funded public health initiatives. To achieve its objectives, GHSC-PSM implemented last-mile distribution in ten regions, utilizing third-party logistics (3PL) transporters to mitigate the high costs associated with vehicle acquisition and fleet management by Regional Medical Stores (RMSs). To promote ongoing improvement, GHSC-PSM conducts bi-annual assessment survey on Last Mile Distribution (LMD) to assess the capacity of 3PL partners in delivering both cold chain and non-cold chain commodities, as well as to verify their delivery claims.

The dashboard was developed to:

- Monitor the performance and capacity of 3PLs across regions.
- Evaluate 3PLs capacity (fleet, temperature control, GPS, etc.).
- Track delivery timelines, cold chain compliance, and on-time delivery rates.
- Cross-check reported deliveries against receiving facility confirmations to flag discrepancies.

Approach

1. **Data Collection & Sources**
    - 3PL Reporting Data:
        - Copies of Proof of Delivery (PoD) submitted electronically and stored on shared drive.
        - Fields: Delivery ID, Route, Commodity Type, Volume, Delivery Date, Cold Chain Requirement, Delivery Status.
    - Verification & 3PL Capacity Assessment Survey Data:

- Bi-annual survey data on sampled facilities and respective 3PL collected via Kobo Toolbox. Kobo Toolbox data connected to Power BI via Kobo API.
- Fields: Facility Name, Date of Last Delivery, Commodity Received, Condition, GPS Tag, Delivery Receipt Signed. 3PL Fleet size, cold storage capacity, compliance with SOPs.

2. **Data Preparation & Modelling**
   - Data cleaned and merged in Power Query.
   - Developed a unified data model:
     - Fact tables (PoD_Assessment, Vendor_Compliance)
     - Dimension tables (VendorList, VehicleSpecs, ModeOfDelivery, LMD_Sites, LMD_Benefits, etc)
   - Created calculated measures using DAX (e.g., calc_InvoiceAvail, Calc_MatchedProducts, calc_NoInvoiceCount, etc)

3. **Dashboard Features**
   - **Cards**: Regions Visited, RMSs Visited, SDPs Visited, PoD Verified, 3PL Assessed.
   - **Visuals**:
     - Interactive map showing locations of regional medical stores (RMS) and healthcare facilities (SDP) visited in the country during the PoD verification and 3PL capacity assessment.
     - Stacked bar charts showing number of PoDs verified in each region visited.
     - Table showing the percentage of SDPs having their PoDs matching the copies submitted to the central level. This visual is also used to show reasons for unavailable invoices in each region.
     - Line and clustered column chart used to compare how many verified PoDs were accompanied by invoices in each region. This visual is also used to compare how many inspected invoices matched the product items delivered in each region as well as showing the status of bin cards / inventory cards updates for each visited facility in each region comparing to the PoDs verified and invoices inspected.

        - Stacked bar chart showing the 3PLs' performance on on-time delivery and as well as 3PLs' performance on loss/overage instance.
        - 100% Stacked bar chart showing 3PLs' performance on temperature monitoring, vehicle specifications, staff support, and mode of delivery.
        - Donut chart showing the rating of LMD benefits by healthcare facilities interviewed.
    - **Interactivity:**
        - **Slicer**: Filter by Year & Month.

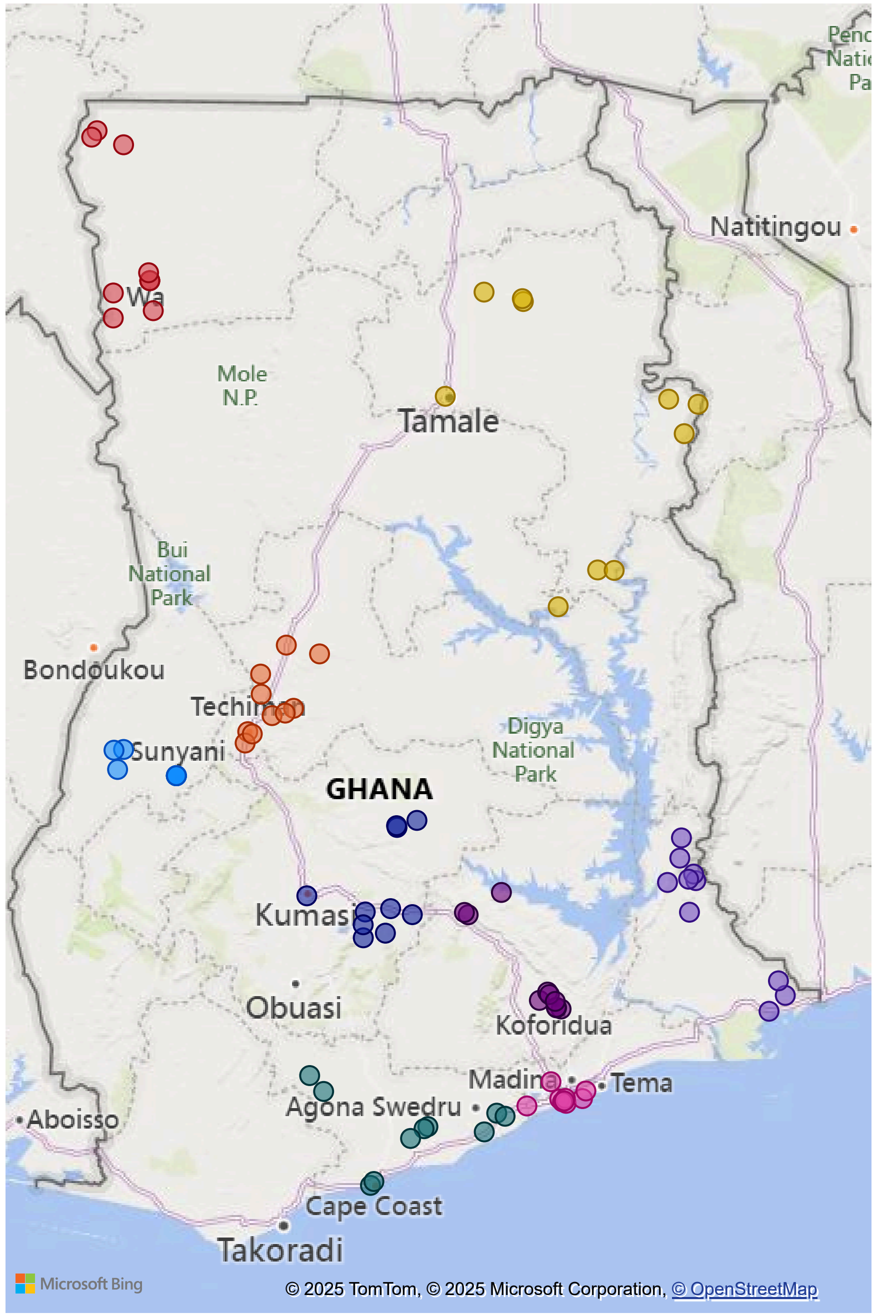
4. **Deployment**
    - Published to Power BI Service.
    - Scheduled data refresh on demand.
    - Shared with M&E Team, Warehousing & Transport Leads and Technical Directors via Power BI service.

Impact

- Improved accountability by validating transporter claims using independent data.
- Enabled route-level performance monitoring, guiding contract renewals and penalties.
- Informed logistics planning by identifying cold chain capacity gaps in hard-to-reach healthcare facilities.
- Strengthened data use culture within the national supply chain coordination unit.

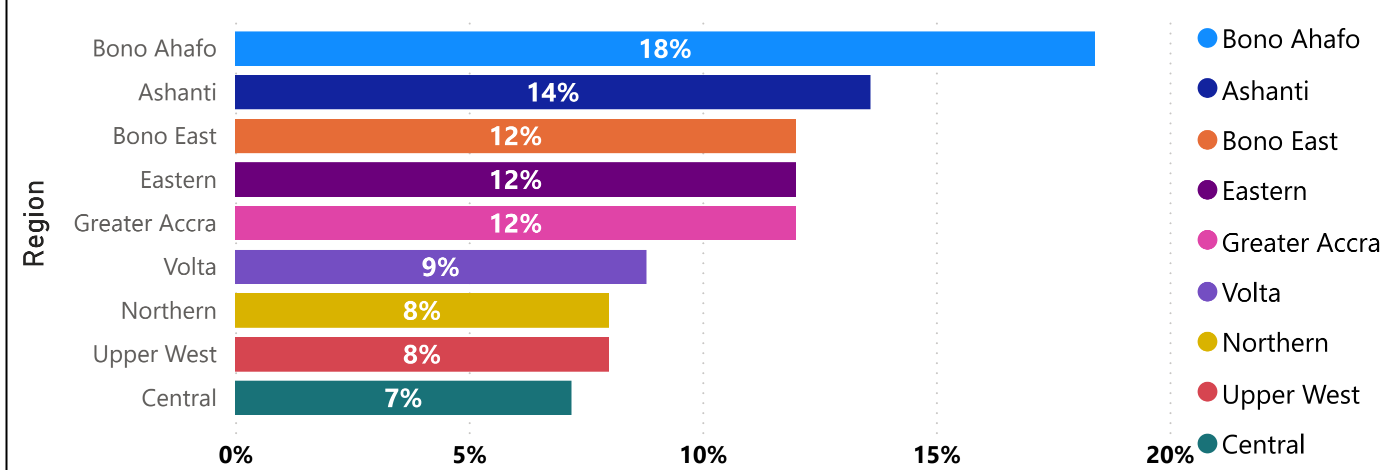
PoD Verification & 3PL Capacity Assessment

RMSs & Healthcare Facilities Visited



9	7	81	125	4
Regions Visited	RMSs Visited	SDPs Visited	PoDs Verified	3PL Assessed

PoD Verified by Region



Select Year & Month

2024

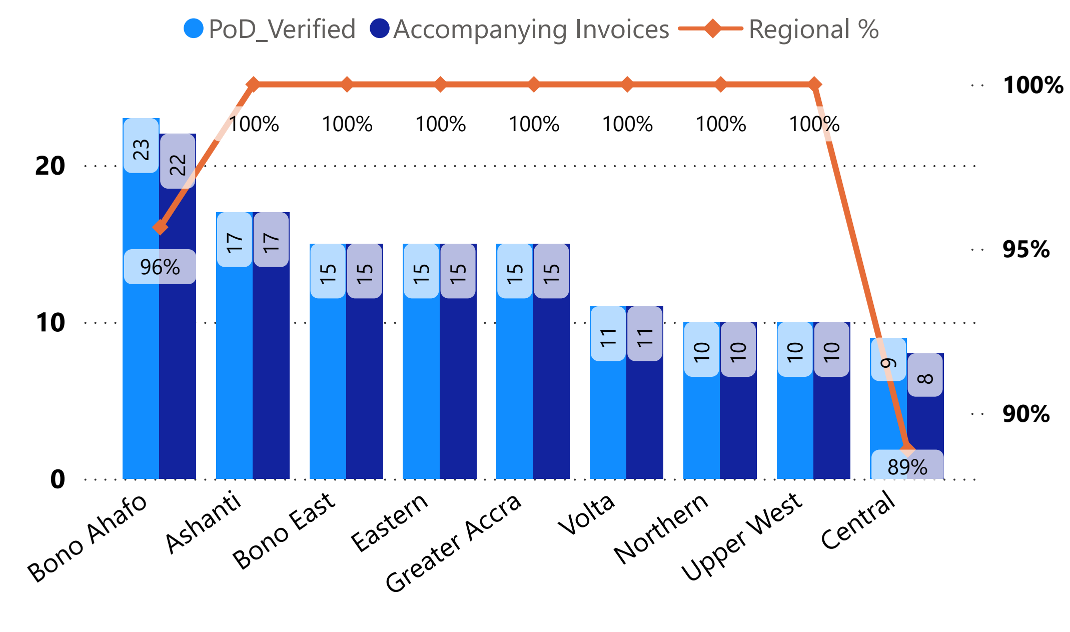
12

Matching SDP's PoD Details To Central Level's Copy

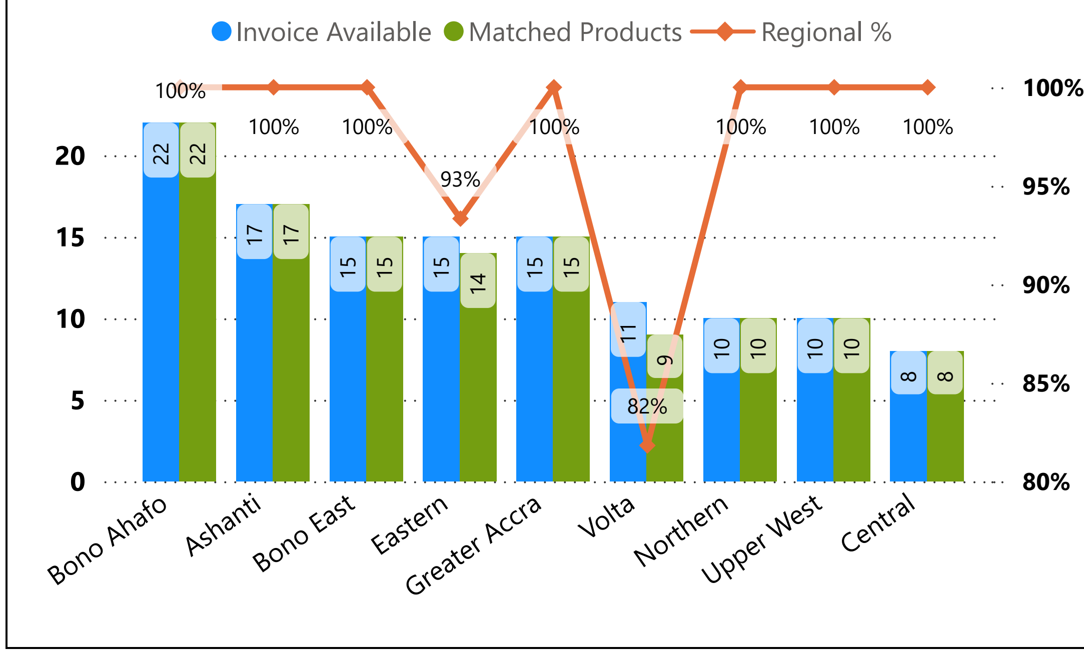
Region	Vendor	Facility Name	Total Packages	Receiver Name	Receiver Signature	Loading Date	Loading Time	Delivery Date	Delivery Time	Regional Total
Ashanti	Sky_Net_Express	100%	76%	100%	100%	76%	76%	76%	76%	85%
Bono Ahafo	Diamond_Logistics	100%	100%	100%	100%	91%	91%	91%	91%	96%
Bono East	Sky_Net_Express	100%	100%	100%	100%	100%	100%	100%	100%	100%
Central	Mass_Logistics	100%	100%	100%	100%	22%	100%	89%	11%	78%
Eastern	Spead_Ocean	100%	100%	100%	100%	100%	100%	100%	100%	100%
Greater Accra	Sky_Net_Express	100%	100%	100%	73%	100%	100%	87%	87%	93%
Northern	Sky_Net_Express	100%	100%	100%	100%	100%	100%	100%	100%	100%
Upper West	Diamond_Logistics	100%	100%	100%	100%	100%	100%	100%	100%	100%
Volta	Mass_Logistics	82%	73%	82%	82%	82%	82%	82%	82%	81%
Total		**98%**	**94%**	**98%**	**95%**	**88%**	**94%**	**91%**	**86%**	**93%**

PoD Verification & 3PL Capacity Assessment

PODs With Accompanying Invoices By Region



Invoices With Matched Products By Region



Select Year & Month

2024

12

Reasons For Unavailable Invoice

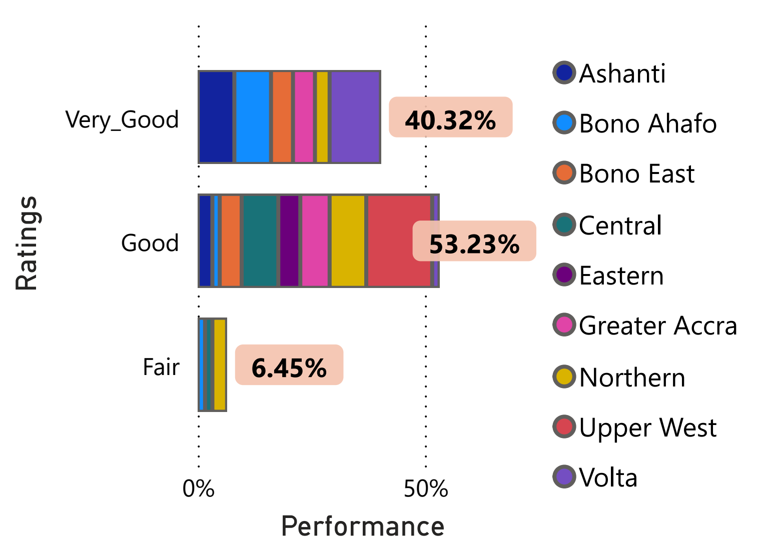
Region	Unavailable Invoice	Reasons For No Invoice	Other Reasons	Regional %
Central	1	Invoice_voucher_kept_at_DHA		11%
Bono Ahafo	1	Other	They couldn't find it	4%
Overall Total	**2**			**2%**

Status of Bin Cards Updates

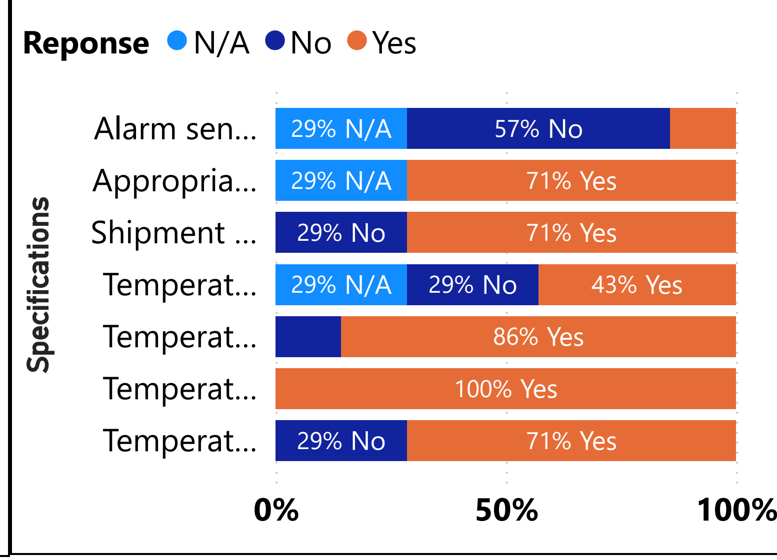
Region	PoD Assessed	Invoice Available	Updated Bin Cards	% Updated
Bono Ahafo	23	22	22	100%
Eastern	15	15	15	100%
Greater Accra	15	15	15	100%
Northern	10	10	10	100%
Upper West	10	10	10	100%
Volta	11	11	11	100%
Central	9	8	6	75%
Bono East	15	15	10	67%
Ashanti	17	17	11	65%
Total	**125**	**123**	**110**	**89%**

PoD Verification & 3PL Capacity Assessment

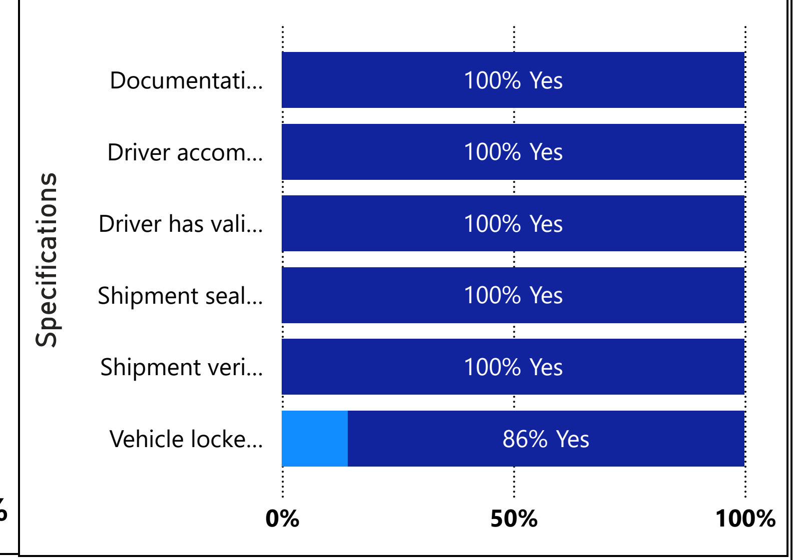
3PL Performance On On-Time Delivery



3PL Performance On Temperature Monitoring



3PL Performance On Staff Support

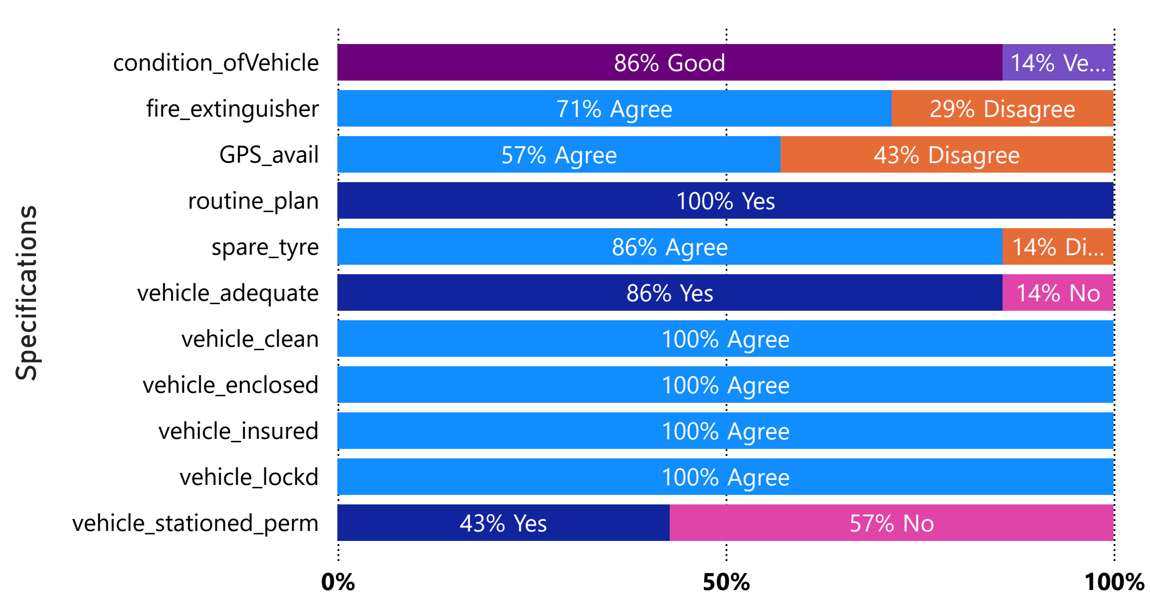


Select Year & Month

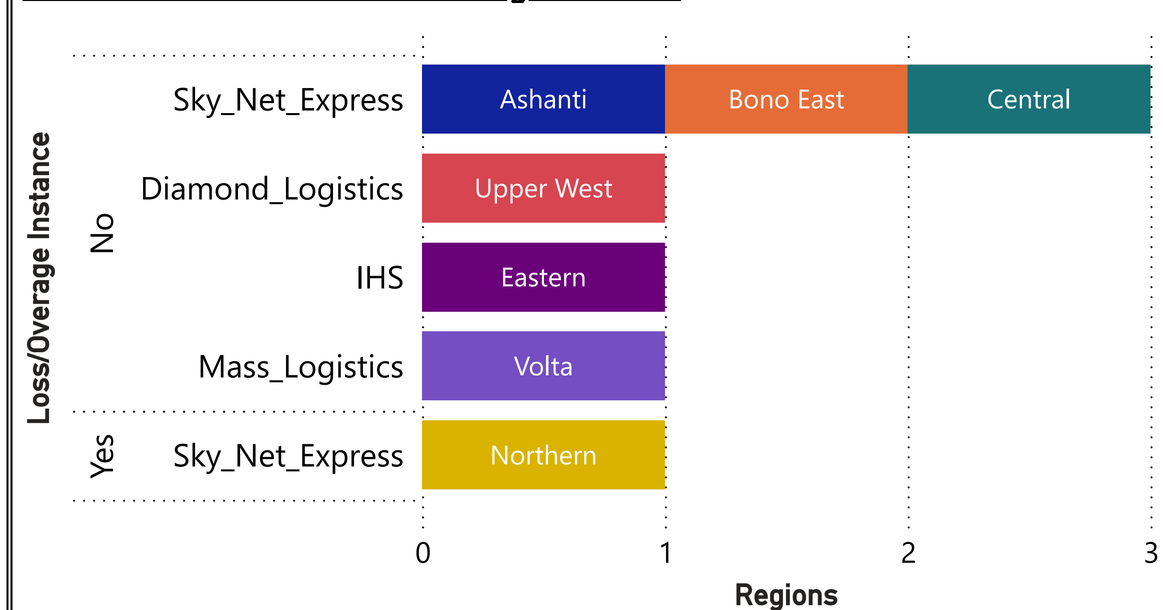
2024

12

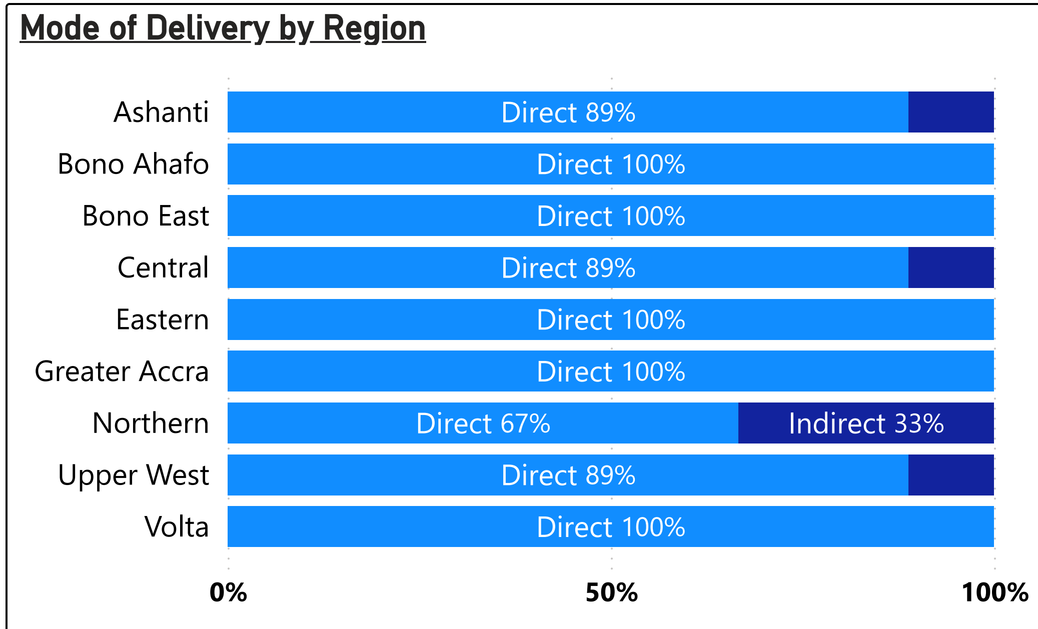
3PL Performance On Vehicle Specifications

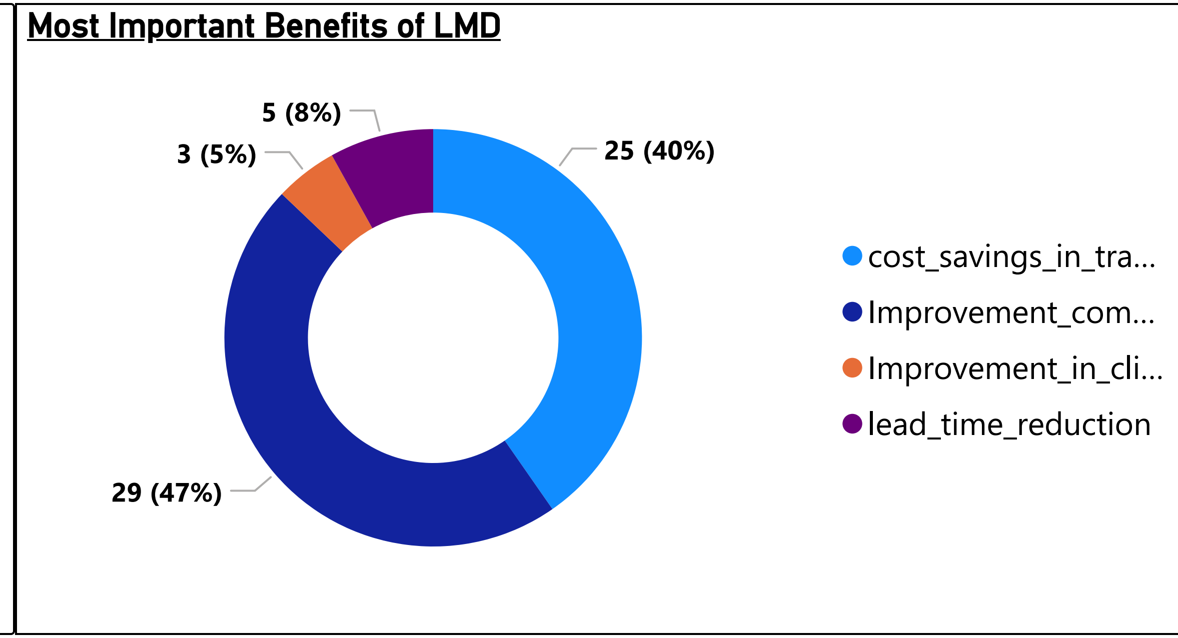


3PL Performance On Loss/Overage Instance



PoD Verification & 3PL Capacity Assessment





Select Year & Month

2024

12

Project: Sales Intelligence Dashboard (SNOWFLAKE)

Background

Snowflake is a cloud-based data platform that offers a comprehensive suite of services, including data warehousing, data lakes, data engineering, data science, and data application development. The platform's scalability and performance enable it to store and efficiently manage large datasets. This capability empowers businesses to make informed, data-driven decisions without concerns about storage limitations or technical obstacles.

The Snowflake sales database is a demonstration database provided by Snowflake that simulates a real-world sales scenario. It is utilized to showcase features, execute practice queries, and explore analytics in contexts such as retail, e-commerce, or B2B sales. This project aims to build an interactive sales intelligence dashboard using the Snowflake sales database to:

- Conduct a comprehensive sales analysis, including cumulative sales, product sales, and profit margins.
- Conduct a cohort analysis to gain insights into customer retention patterns.
- Integrate time-series forecasting to predict future profits using historical data.

The Snowflake Sales Database file is substantial, measuring just under 500 megabytes. To optimize the performance of Power BI Desktop—ensuring smoother loading, efficient data refresh, and overall responsiveness—a high-end PC with a minimum of 16 gigabytes of RAM is recommended. Additionally, the data within the database table has been obfuscated to safeguard the original nature of the information.

Approach

1. **Data Sources**
    - Connected Power BI to Snowflake using the ODBC connector and imported normalized tables:
        - Calendar, Customer, Product, Sales, Store

2. **Data Preparation & Modelling**
    - Created a star schema with Sales as the fact at the centre. Dimension tables: Store, Product, Customer, Calendar
    - Added calculated measure in DAX for:
        - Total Sales, Total Quantity, Total Profits, Total Profit Margins, Cohort Average Sales, Cumulative Sales, etc.

3. **Dashboard Features**
    - **Interactivity:**
        - Slicer for time period where you can select a year, a quarter and a month. There are also slicers for customers, store, store city, product category, and manufacturer.
    - **Visuals**:
        - Area chart showing cumulative sales as well as cumulative sales last year of whatever is selected in the slicers.
        - Table showing city, product name, sales, profits, profit margins, and total quantity.
        - Clustered bar chart showing product and product sales.
        - Clustered column chart showing total profit by year and month based whatever store, city, or customer selected.
        - Cards showing first sale date, last sale date, total sales, sales last year, total profits, and average sales per customer.
        - Matrix displaying cohort analysis of customer purchasing patterns for a specific manufacturer or product category selected.
        - Line chart showing total profit by dates based on the selected store as well as total profit projection for the next 90 days.
4. **Deployment**
    - Published to Power BI Service.

<u>Impact</u>

- Enabled quarterly planning using forecasts with >85% accuracy.
- Helped optimize inventory and marketing campaigns by identifying top-converting cohorts.
- Improved sales performance tracking across departments and channels.
- Increased executive visibility with self-service analytics through Power BI Service.

Time Period

All

Customers

Search

- Select all
- (Blank)
- Aaron
- Abbott
- Abel
- Abell
- Abernathy
- Abner
- Abney
- Abraham
- Abrams
- Abreu

Store

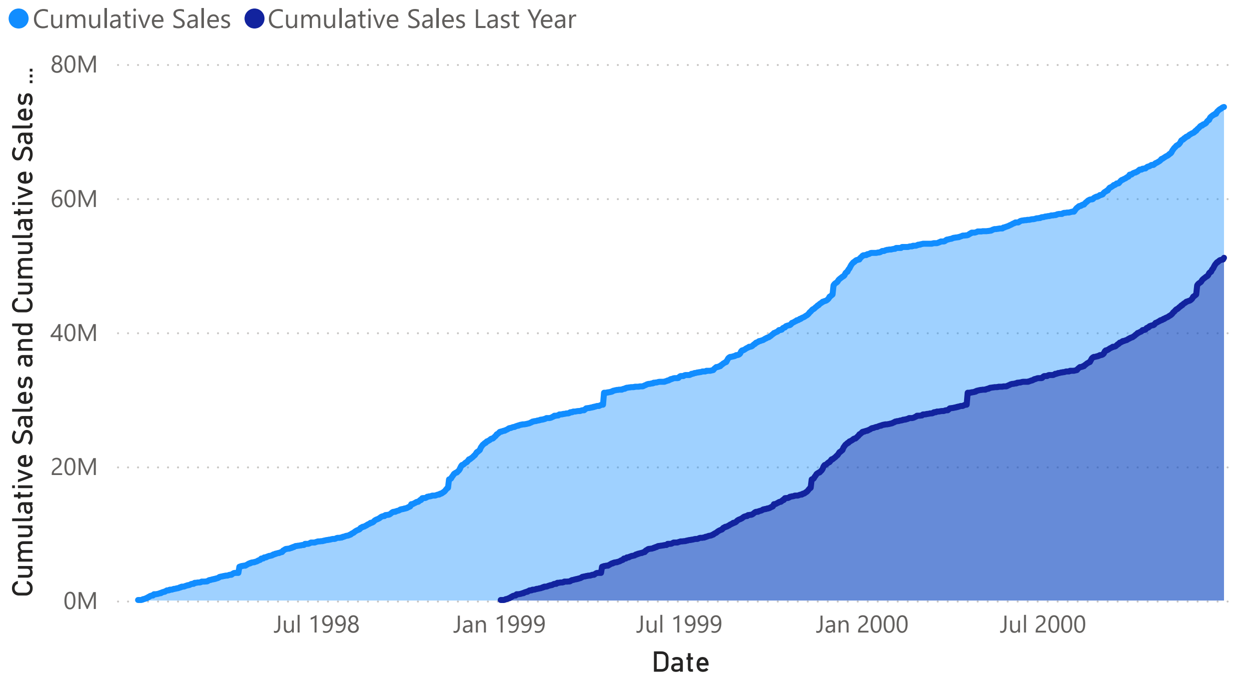
Search

- (Blank)
- Store: 1
- Store: 10
- Store: 100
- Store: 1000
- Store: 1003
- Store: 1004

Store City

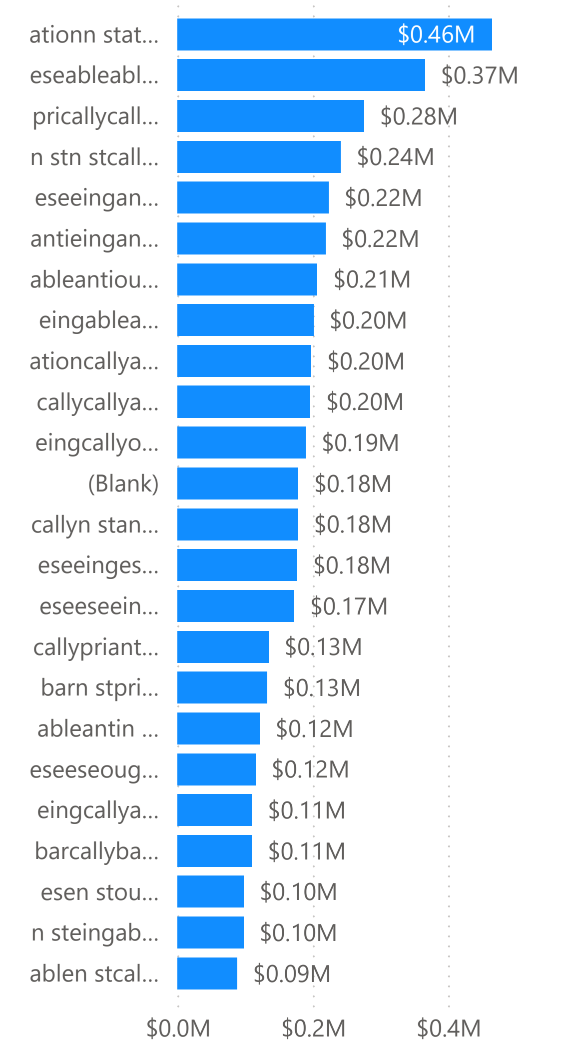
- Oak Hill

Cumulative Sales



City	Product Name	Sales	Profits	Profit Margin	Total Quantity
Oak Hill		$178,228	$60,892	34.17%	2,252
Oak Hill	ableableableableantiought	$5,558	$2,553	45.94%	49
Oak Hill	ableableableablen stpri	$6,368	$2,888	45.35%	56
Oak Hill	ableableableableoughtable	$2,456	$714	29.08%	79
Oak Hill	ableableableablepri	$3,378	$1,663	49.24%	19
Oak Hill	ableableableantieseought	$1,063	$139	13.03%	32
Oak Hill	ableableableationought	$7,101	$273	3.84%	72
Total		**$73,593,023**	**$24,265,938**	**32.97%**	**987,336**

Product Sales



03/01/1998

First Sale Date

31/12/2000

Last Sale Date

$73,593,023

Total Sales

$51,104,891

Sales Last Year

$24,265,938

Total Profits

69.44%

Sales Growth

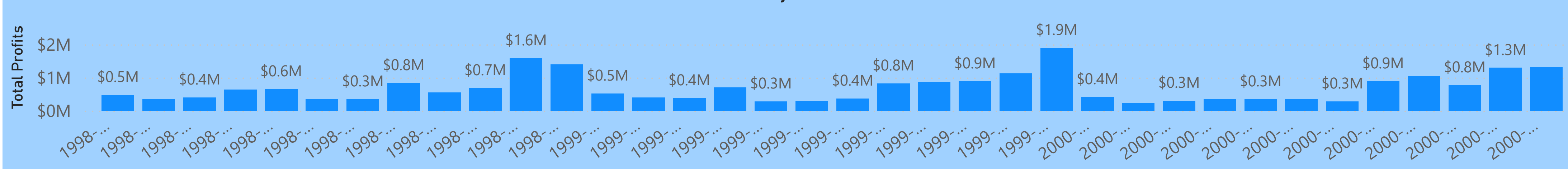
1,652

Customer Count

$44,548

Avg. Sale Per Customer

Total Profits by Year and Month



Category

- Books
- Children
- Electronics
- Home
- Jewelry
- Men
- Music
- Shoes
- Sports
- Women

Manufacturer

- (Blank)
- able
- ableable
- ableableable
- ableableanti
- ableableation
- ableablecally
- ableableeing
- ableableese
- ableablen st
- ableableought
- ableablepri
- ableanti
- ableantiable
- ableantianti
- ableantiation
- ableantically
- ableantieing
- ableantiese
- ableantin st
- ableantiought
- ableantipri

Cohort	0	1	2	3	4	5	6	7	8	9	10	11	12
1998-01	$42,605	$41,775	$41,986	$41,709	$42,183	$42,054	$42,118	$41,921	$42,212	$42,337	$42,215	$42,267	$42,323
1998-02	$41,775	$41,986	$41,709	$42,183	$42,054	$42,118	$41,921	$42,212	$42,337	$42,215	$42,267	$42,323	$42,720
1998-03	$41,986	$41,709	$42,183	$42,054	$42,118	$41,921	$42,212	$42,337	$42,215	$42,267	$42,323	$42,720	$42,190
1998-04	$41,709	$42,183	$42,054	$42,118	$41,921	$42,212	$42,337	$42,215	$42,267	$42,323	$42,720	$42,190	$42,157
1998-05	$42,183	$42,054	$42,118	$41,921	$42,212	$42,337	$42,215	$42,267	$42,323	$42,720	$42,190	$42,157	$42,390
1998-06	$42,054	$42,118	$41,921	$42,212	$42,337	$42,215	$42,267	$42,323	$42,720	$42,190	$42,157	$42,390	$42,170
1998-07	$42,118	$41,921	$42,212	$42,337	$42,215	$42,267	$42,323	$42,720	$42,190	$42,157	$42,390	$42,170	$42,296
1998-08	$41,921	$42,212	$42,337	$42,215	$42,267	$42,323	$42,720	$42,190	$42,157	$42,390	$42,170	$42,296	$42,104
1998-09	$42,212	$42,337	$42,215	$42,267	$42,323	$42,720	$42,190	$42,157	$42,390	$42,170	$42,296	$42,104	$42,245
1998-10	$42,337	$42,215	$42,267	$42,323	$42,720	$42,190	$42,157	$42,390	$42,170	$42,296	$42,104	$42,245	$42,354
1998-11	$42,215	$42,267	$42,323	$42,720	$42,190	$42,157	$42,390	$42,170	$42,296	$42,104	$42,245	$42,354	$42,385
1998-12	$42,267	$42,323	$42,720	$42,190	$42,157	$42,390	$42,170	$42,296	$42,104	$42,245	$42,354	$42,385	$42,129
1999-01	$42,323	$42,720	$42,190	$42,157	$42,390	$42,170	$42,296	$42,104	$42,245	$42,354	$42,385	$42,129	$42,622
1999-02	$42,720	$42,190	$42,157	$42,390	$42,170	$42,296	$42,104	$42,245	$42,354	$42,385	$42,129	$42,622	$42,432
1999-03	$42,190	$42,157	$42,390	$42,170	$42,296	$42,104	$42,245	$42,354	$42,385	$42,129	$42,622	$42,432	$42,121
1999-04	$42,157	$42,390	$42,170	$42,296	$42,104	$42,245	$42,354	$42,385	$42,129	$42,622	$42,432	$42,121	$42,004
1999-05	$42,390	$42,170	$42,296	$42,104	$42,245	$42,354	$42,385	$42,129	$42,622	$42,432	$42,121	$42,004	$42,088
1999-06	$42,170	$42,296	$42,104	$42,245	$42,354	$42,385	$42,129	$42,622	$42,432	$42,121	$42,004	$42,088	$42,078
1999-07	$42,296	$42,104	$42,245	$42,354	$42,385	$42,129	$42,622	$42,432	$42,121	$42,004	$42,088	$42,078	$42,304
1999-08	$42,104	$42,245	$42,354	$42,385	$42,129	$42,622	$42,432	$42,121	$42,004	$42,088	$42,078	$42,304	$42,391
1999-09	$42,245	$42,354	$42,385	$42,129	$42,622	$42,432	$42,121	$42,004	$42,088	$42,078	$42,304	$42,391	$42,267
1999-10	$42,354	$42,385	$42,129	$42,622	$42,432	$42,121	$42,004	$42,088	$42,078	$42,304	$42,391	$42,267	$42,565
1999-11	$42,385	$42,129	$42,622	$42,432	$42,121	$42,004	$42,088	$42,078	$42,304	$42,391	$42,267	$42,565	$42,409
1999-12	$42,129	$42,622	$42,432	$42,121	$42,004	$42,088	$42,078	$42,304	$42,391	$42,267	$42,565	$42,409	$42,128
2000-01	$42,622	$42,432	$42,121	$42,004	$42,088	$42,078	$42,304	$42,391	$42,267	$42,565	$42,409	$42,128	
2000-02	$42,432	$42,121	$42,004	$42,088	$42,078	$42,304	$42,391	$42,267	$42,565	$42,409	$42,128		
2000-03	$42,121	$42,004	$42,088	$42,078	$42,304	$42,391	$42,267	$42,565	$42,409	$42,128			
2000-04	$42,004	$42,088	$42,078	$42,304	$42,391	$42,267	$42,565	$42,409	$42,128				
2000-05	$42,088	$42,078	$42,304	$42,391	$42,267	$42,565	$42,409	$42,128					
2000-06	$42,078	$42,304	$42,391	$42,267	$42,565	$42,409	$42,128						
2000-07	$42,304	$42,391	$42,267	$42,565	$42,409	$42,128							
2000-08	$42,391	$42,267	$42,565	$42,409	$42,128								
2000-09	$42,267	$42,565	$42,409	$42,128									
2000-10	$42,565	$42,409	$42,128										

Store Name

- (Blank)
- Store: 1
- Store: 10
- Store: 100
- Store: 1000
- Store: 1003
- Store: 1004
- Store: 1006
- Store: 1009
- Store: 1010
- Store: 1012
- Store: 1015
- Store: 1016
- Store: 1018
- Store: 1021
- Store: 1022
- Store: 1024
- Store: 1027
- Store: 1028
- Store: 103
- Store: 1030
- Store: 1033
- Store: 1034
- Store: 1036
- Store: 1039
- Store: 104
- Store: 1040
- Store: 1042
- Store: 1045